Probabilistic Language Modeling

Motivation and Estimation

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آکادمیراه

Outline

- **Motivation**
- Estimation

Language Modeling

- Finding the probability of a sentence or a sequence of words

$$P(S) = P(w_1, w_2, w_3, w_4, w_5, \dots, w_n)$$

- Applications:
  - Word prediction
  - Speech recognition
  - Machine translation
  - Spell checker

Applications

- Word Prediction

"natural language" → "processing"
"management"

Applications

- Speech recognition

"Computers can recognize speech."

"Computers can wreck a nice peach."

Applications

- Machine translation

"The cat eats ..." → "Die Katze frisst ..."

"Die Katze isst ..."

Applications

- Spell checker

"I want to adver this project." → "advert" / "adverb"

Outline

- Motivation
- **Estimation**

Corpus

- Probabilities are based on counting things
- Counting of thing in natural language is based on a corpus (plural: corpora)
- A computer-readable collection of text or speech
  - The Brown Corpus
    - A million-word collection of samples
    - 500 written texts from different genres
    - (newspaper, fiction, non-fiction, academic, ...)
    - Assembled at Brown University in 1963-1964
  - The Switchboard Corpus
    - A collection of 240 hours of telephony conversations
    - 3 million words in 2430 conversations averaging 6 minutes each
    - Collected in early 1990s

Text Corpora I

- American National Corpus
- Bank of English
- British National Corpus
- Bergen Corpus of London Teenage Language (COLT)
- Brown Corpus, forming part of the "Brown Family" of corpora, together with LOB, Frown and F-LOB
- Corpus of Contemporary American English (COCA) 425 million words, 1990–2011. Freely searchable online
- Corpus Resource Database (CoRD), more than 80 English language corpora.

Text Corpora II

- Coruña Corpus, a corpus of late Modern English scientific writing covering the period 1700-1900, developed by the Muste research group at the University of A Coruña
- GUM corpus, the open source Georgetown University Multilayer corpus, with very many annotation layers
- Google Books Ngram Corpus
- International Corpus of English
- Oxford English Corpus
- RE3D (Relationship and Entity Extraction Evaluation Dataset)
- Santa Barbara Corpus of Spoken American English
- Scottish Corpus of Texts & Speech

Text Corpora III

- Apache Software Foundation Public Mail Archives: all publicly available Apache Software Foundation mail archives as of July 11, 2011 (200 GB)
- Blog Authorship Corpus: consists of the collected posts of 19,320 bloggers gathered from blogger.com in August 2004. 681,288 posts and over 140 million words. (298 MB)
- Amazon Fine Food Reviews [Kaggle]: consists of 568,454 food reviews Amazon users left up to October 2012. Paper. (240 MB)
- Amazon Reviews: Stanford collection of 35 million amazon reviews. (11 GB)
- ArXiv: All the Papers on archive as fulltext (270 GB) + sourcefiles (190 GB).

Text Corpora IV

- CLiPS Stylometry Investigation (CSI) Corpus: a yearly expanded corpus of student texts in two genres: essays and reviews. The purpose of this corpus lies primarily in stylometric research, but other applications are possible. (on request)
- ClueWeb09 FACC: ClueWeb09 with Freebase annotations (72 GB)
- ClueWeb11 FACC: ClueWeb11 with Freebase annotations (92 GB)
- Common Crawl Corpus: web crawl data composed of over 5 billion web pages (541 TB)
- Cornell Movie Dialog Corpus: contains a large metadata-rich collection of fictional conversations extracted from raw movie scripts: 220,579 conversational exchanges between 10,292 pairs of movie characters, 617 movies (9.5 MB)

Word Occurrence

- A language consist of a set of V words (Vocabulary)
- A text is a sequence of the words from the vocabulary
- A word can occur several times in a text
  - Word Token: each occurrence of words in text
  - Word Type: each unique occurrence of words in the text

Word Occurrence

Example:

This is a sample text from a book that is read every day

Word Occurrence

Example:

This is a sample text from a book that is read every day

Word Tokens: 13

Word Types: 11

Counting

- Brown
  - 1,015,945 word tokens
  - 47,218 word types
- Google N-gram
  - 1,024,908,267,229 word tokens
  - 13,588,391 word types

That seems like a lot of types...

Even large dictionaries of English have only around 500k types.

Why so many here?

Numbers

Misspellings

Names

Acronyms

Language Modeling

- Finding the probability of a sentence or a sequence of words

$$P(S) = P(w_1, w_2, w_3, w_4, w_5, \ldots, w_n)$$

P(Computer, can, recognize, speech)

Bayes Decomposition

- Write joint probability as product of conditional probabilities

$$P(w_1 ; w_2) = P(w_1) \cdot P(w_2 | w_1)$$

$$P(w_1,w_2,w_3,w_4) = P(w_1) \cdot P(w_2 | w_1) \cdot P(w_3 | w_1,w_2) \cdot P(w_4 |w_1,w_2,w_3)$$

$$P(w_1,w_2,\dots , w_n) = P(w_1) \cdot P(w_2 | w_1) \cdot P(w_3 | w_1,w_2) \cdots P(w_n |w_1,w_2,\dots , w_{n-1})$$

$$P(S) = (w_1) \cdot P(w_2 | w_1) \cdot P(w_3 | w_1,w_2) \cdots P(w_n |w_1,w_2,\dots , w_{n-1})$$

$$P(S) = \prod_{i=1}^{n} P(w_i |w_1,w_2,\dots , w_{i-1})$$

Conditional Probability

$$P(S) = \prod_{i=1}^{n} P(w_i \mid w_1, w_2, \ldots , wi_{-1})$$

P(Computer, can, recognize, speech) =

P(Computer)· P(can|Computer)·

P(recognize|Computer,can)·

P(speech|Computer,can,recognize)

Maximum Likelihood Estimation

P(speech | Computer can recognize)

$$P(\text{speech}|\text{Computer can recognize}) = \frac{\#(\text{Computer can recognize speech})}{\#(\text{Computer can recognize})}$$

- Too many phrases
- Limited text for estimating the probability
  - Making a simplification assumption

Markov Assumption

$$P(S) = \prod_{i=1}^{n} P(w_i \mid w_1, w_2, \ldots, w_{i-1})$$

$$P(S) = \prod_{i=1}^{n} P(w_i \mid w_{i-1})$$

Andrei Markov

P(Computer; can; recognize; speech) =

P(Computer) · P(can|Computer) · P(recognize|can) · P(speech|recognize)

$$(speech|recognize) = \frac{\#(\text{recognize speech})}{\#(\text{recognize})}$$

N-gram Model

- **Unigram** $P(S) = \prod_{i=1}^{n} P(w_i)$
- **Bigram** $P(S) = \prod_{i=1}^{n} P(w_i \mid w_{i-1})$
- **Trigram** $P(S) = \prod_{i=1}^{n} P(w_i \mid w_{i-2} w_{i-1})$
- **N-gram** $P(S) = \prod_{i=1}^{n} P(w_i \mid w_1 w_2 , \ldots , w_{i-1})$

Maximum Likelihood

<s> I saw the boy </s>

<s> the man is working </s>

<s> I walked in the street </s>

Vocab:

I saw the boy man is working walked in street

boy I in is man saw street the walked working

Maximum Likelihood

<s> I saw the boy </s>

<s> the man is working </s>

<s> I walked in the street </s>

boy	I	in	is	Man	saw	street	The	walked	working
1	2	1	1	1	1	1	3	1	1

Maximum Likelihood

boy	I	in	is	Man	saw	street	The	walked	working
1	2	1	1	1	1	1	3	1	1

	boy	I	in	is	Man	saw	street	The	walked	working
boy										
I						1			1	
in								1		
is										1
Man				1						
saw								1		
street										
The	1				1		1			
walked			1							
working										

Maximum Likelihood

<s> I saw the man </s>

$$P(S) = P(I) \cdot P(saw|I) \cdot P(the|saw) \cdot P(man|the)$$

$$P(S) = \frac{\#(I)}{\#} \quad \frac{\#(I\ saw)}{\#(I)} \quad \frac{\#(saw\ the)}{\#(saw)} \quad \frac{\#(the\ man)}{\#(the)}$$

$$P(S) = \frac{2}{13} \quad \frac{1}{2} \quad \frac{1}{1} \quad \frac{1}{3}$$

Further Reading

- Speech and Language Processing (3rd ed. draft)
  - Chapter 3